Capacitively-Coupled Chopper Instrumentation Amplifier for Linear Hall Effect Sensor

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1 Introduction

[1]

- Discuss magnetic sensing application space
  - Current sensing
  - Tactile Sensing (robotics)
  - Position/Velocity Sensing (automotive)
  - Biosensors
- Discuss utility of building open-source Hall sensor readout
  - No existing data on n-well-based Hall sensors in open PDKs
  - Sensor evaluation gives future designers ability to integrate into their application
  - Readout circuit directly applicable to most bridge-type sensors

2 Conceptual Design

- Show high-level block diagram
- Table of pinouts with signal descriptions/directions

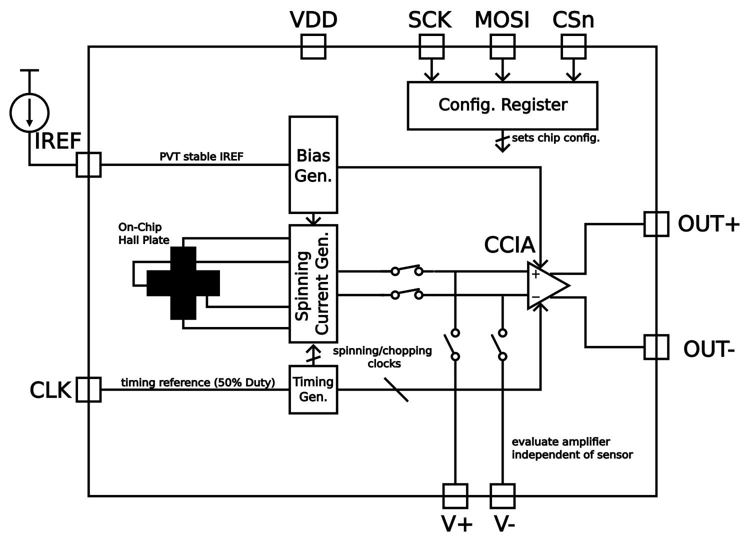

Figure 1: Block diagram of the proposed architecture.

Fig. 1
Table 1

2.1 Sensor

- Vertical vs. Horizontal sensor design
- Horizontal sensor geometries
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2.2 Readout Circuit

- Instrumentation Amplifier options/survey
- CCIA chosen due to large common-mode input range (sweepable bias current), stable gain (easy to integrate several gain options if needed)
- Hall sensor bias expected to dominate power consumption, but CCIA being low-power is great in other high-impedance sensor applications
- analog output reduces complexity and allows for easier characterization of CCIA

Table 1: Chip pinout description.

Analog Pins		
IREF	Analog	Current reference
OUT+	Analog	Differential output positive terminal
OUT-	Analog	Differential output negative terminal
VTP+	Analog	Positive analog test point (differential probe node)
VTP-	Analog	Negative analog test point (differential probe node)
Digital Interface		
SCK	Digital Input	SPI configuration clock
MOSI	Digital Input	SPI configuration data input
CSn	Digital Input	SPI chip select
CLK	Digital Input	Chopping/spinning clock reference
Supply Pins		
VDD	Power	Supply input
VSS	Power	Ground (shared with other Chipathon blocks)

3 Design Specifications

- Table of specifications for full system

3.1 Sensor

3.2 Readout Circuit

4 Schematic Design

- Show top-level schematic

4.1 Subblock 1

4.2 Subblock N

5 Layout Design

- Show layout floorplan

5.1 Subblock 1

5.2 Subblock N

6 Simulation Results

- Verification Plan details (how is each specification tested?)

6.1 Simulation 1

6.2 Simulation M

References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.